Group 1

Lab 10 – Unsupervised Learning

ITAI-1371

Viswanatha Rao

Team Member Contribution Journal

Viktoriya Kurmisheva:

Worked on Task 2 in Part 2, which focused on performing K-Means clustering using four clusters and creating a scatter plot to visualize the data points based on their assigned clusters. Answered Question 2 in the Knowledge Check section by explaining why choosing the largest possible number of clusters is not always the best approach, even though it produces lower inertia. After completing the lab, gathered the group's thoughts and observations about the activities, clustering results, PCA visualization, and concepts covered in the notebook. Used this information to write and organize the group's Reflection Journal, then uploaded the completed journal to the group's GitHub repository.

Tien Manh Nguyen:

Worked on Task 1 in Part 2, which involved applying the Elbow Method to calculate inertia for different numbers of clusters and creating a plot to help identify the optimal value of `k`. Answered Question 1 in the Knowledge Check section by explaining the fundamental difference between supervised and unsupervised learning. After the group completed their assigned tasks, reviewed the notebook's code to check for errors and make sure the clustering and PCA implementations worked correctly. Then uploaded the completed Jupyter Notebook to the group's GitHub repository.

Zaid Alayedy:

Worked on Task 3 in Part 3, which focused on applying Principal Component Analysis (PCA) to reduce the Iris dataset from four dimensions to two principal components and visualizing the results. Answered Question 3 in the Knowledge Check section by explaining what it means when the first two principal components preserve over 95% of the variance in the original dataset. In addition, gathered the contribution details from each group member and used them to write and organize the Team Member Contribution Journal. Reviewed the descriptions to make sure everyone's responsibilities were accurately represented, then uploaded the completed journal to the group's GitHub repository.